ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet
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Date: 5/1/2025
Compounds: Amoxicillin, Aspirin, Bismuth Subsalicylate, Caffeine

New Compound Analysis

Name: AmoxoSalicaffeine
Formula: $C_{20}H_{22}N_4O_8S$
Weight: 474.48
Drug-likeness: 0.76

Activity:
Antibacterial and Analgesic

Target Analysis

Penicillin-Binding Protein
Binding Affinity: -9.5
Critical for bacterial cell wall synthesis inhibition

Cyclooxygenase (COX-1 and COX-2)
Binding Affinity: -8.1
Important for anti-inflammatory and analgesic effects

Toxicity Profile

Risk Level: Moderate
Affected Systems: Gastrointestinal tract, Kidneys, Liver

Mitigation Suggestions:
- Use of protective agents like proton pump inhibitors
- Regular kidney function tests
- Monitoring liver enzymes

Pharmacokinetics

Absorption:
Moderate oral bioavailability

Distribution:
Moderate Vd with serum protein binding

Metabolism:
Hepatic, involves CYP450 enzymes

Excretion:
Renal and fecal pathways

Half-life: 5-7 hours

Detailed Analysis

AmoxoSalicaffeine is a novel compound combining the properties of amoxicillin, aspirin, and caffeine, aiming to minimize infection-associated pain and inflammation while maintaining antibiotic efficacy. It provides dual action by simultaneously disrupting bacterial cell wall synthesis and inhibiting cyclooxygenase enzymes. The CNS stimulatory effects from caffeine may counter sedative side effects, ensuring patient alertness. However, potential adverse effects include gastrointestinal irritation and enhanced renal toxicity, necessitating careful dose management and supportive care options such as PPIs for protection. Optimization of pharmacokinetics through improved formulation and delivery forms part of the ongoing research, as does expanding preclinical and clinical trials to confirm efficacy and safety.
